

Dual Spatio-Temporal Contrastive Learning Network With Adaptive Threshold Generation for Anomaly Detection of Electric Submersible Pump

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Abstract—To improve the electric submersible pump (ESP) system's anomaly monitoring performance, this article proposes a novel approach known as the dual spatio-temporal contrastive learning network with adaptive threshold generation (DSTCL-ATG). Unlike previous ESP process modeling methods, this study comprehensively considers the spatio-temporal coupling characteristics of ESP data and incorporates Crossformer into the dual-path contrastive learning (DCL) architecture to provide superior normal ESP process modeling. Furthermore, we design an ATG approach based on a random forest regressor that is aimed at successfully mitigating frequent false alarms resulting from fluctuations in ESP status. The algorithm is evaluated using data from four faulty wells in real oilfield scenarios, demonstrating its effectiveness and superiority through extensive comparative experiments against state-of-the-art methodologies.

Index Terms—Anomaly detection, contrastive learning, electric submersible pump (ESP), process modeling, spatio-temporal feature extraction.

I. INTRODUCTION

IN RECENT years, there has been a notable shift toward artificial lifting as the primary method for oil production. This transition is attributed to the depletion of easily accessible reservoirs and the deterioration of oil well conditions

[1]. As an important artificial lifting equipment, the electric submersible pump (ESP) is extensively employed in various well types, including deep wells, directional wells, and those containing water, due to its strong lifting capacity, large displacement, and high lift [2]. However, ensuring the safe and reliable operation of ESPs poses significant challenges. These challenges stem from their intricate design, complex coupling mechanisms, and continuous operation in harsh oil well environments characterized by high temperatures, pressures, impurities, and corrosion. Like most industrial systems [3], [4], the failure of an ESP could trigger a cascade of events leading to prolonged shutdowns, resulting in substantial economic losses and potentially catastrophic outcomes [5]. Therefore, it is essential to swiftly detect any irregularities in ESP operation to maintain the safety and reliability of the oil recovery process. This necessitates the implementation of effective monitoring strategies to mitigate risks and ensure the uninterrupted operation of ESPs in oil production [6].

The initial focus of many studies has been on two key areas: current card detection [7], [8] and wellhead pressure-holding diagnosis [9], [10]. The current card detection method involves assessing the fluctuation trend of the ESP's current, typically relying on the subjective judgment of field engineers. While this method is straightforward and user-friendly, achieving accurate detection is often challenging due to its subjective assessment criteria [11]. On the other hand, the wellhead pressure-holding diagnosis approach involves increasing pipeline pressure by closing the production valve and then observing pressure changes to identify faults. However, this method is associated with prolonged downtime between stopping and resuming the pump, significantly disrupting regular production operations [12]. Both methods suffer from limitations, as they rely solely on a single variable (such as current or pressure) for ESP abnormality detection. This reliance may lead to low diagnostic accuracy and efficiency, hampering the effectiveness of anomaly detection efforts [13].

With the rapid development of sensing technologies and the vast accumulation of big data, modern approaches such as multivariate statistical analysis and machine learning have emerged as powerful tools for ESP abnormality detection [6], [14], [15]. For instance, classic machine learning algorithms, such as support vector machine (SVM), K -nearest neighbor (KNN), neural networks (NNs), and classification and regres-

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sion trees (CART), have been utilized to detect ESP anomalies. Despite their encouraging outcomes, these methods primarily concentrate on capturing the multivariate and nonlinear aspects of ESP data, ignoring its dynamics. Besides, they often require pre-labeling of the data, which may limit their applicability in real-world scenarios.

In practice, a significant portion of the recorded ESP data lacks labels, which has led to growing interest in unsupervised detection methods [16], [17], [18], [19]. In [16], principal component analysis (PCA) is utilized to identify anomalies related to ESP broken shafts. In [17], a method combining PCA with individual X and moving range (XmR) is developed for ESP anomaly detection. In [18], an autoencoder (AE) is employed to detect ESP anomalies. Despite the success of these methods, they tend to overlook the temporal dependencies among monitoring variables. Recently, a novel approach known as dynamic kernel PCA with local preservation projection (LDKPCA) has been introduced for the detection of shaft breakage [19]. This method integrates dynamic analysis to effectively capture temporal correlations among variables. Nevertheless, LDKPCA does not serve as an enhancement to NNs and is unable to leverage the powerful feature extraction capabilities inherent in NNs.

Although a lot of research has been done on NN-based dynamic modeling, such as recurrent NNs (RNNs) [20], gated recurrent unit (GRU) networks [21], long short-term memory (LSTM) networks [22], these methods still struggle to learn long-term dependency among variables. Notably, our team developed a temporal attention network (TAN) based on an LSTM-AE architecture, which has demonstrated that NN-based dynamic modeling can be effectively applied to ESP anomaly detection [23]. Nevertheless, the fundamental limitation in capturing complex long-term variable correlations persists across these methodologies. Fortunately, Transformer introduces a multihead attention and residual learning mechanism to improve the long-term dependency modeling performance [24]. However, Transformer and most of its variants focus more on relationship learning in the temporal dimensions while ignoring explicit correlation modeling in the spatial (variable) dimensions [25]. In fact, along with the correlation between variables that share the same physical characteristics, the ESP system also finds a link between neighboring variables with different properties in space. More recently, Crossformer [26] has been proposed and demonstrated to be capable of explicitly modeling spatio-temporal dependency; however, it adopts the classical supervised learning paradigm, which is difficult to be directly used in the unsupervised normal process modeling of ESP anomaly detection task. In the past few years, contrastive learning [27] has achieved sufficient performance in dealing with unsupervised problems due to its outstanding contrastive optimization strategy, which has also been successfully applied in anomaly detection [28], [29]. Nevertheless, these generic techniques fail to account for the spatio-temporal interaction of ESP variables. To our knowledge, there are rarely published reports of unsupervised modeling of ESP systems with spatio-temporal interaction of variables using contrastive learning. Therefore, conducting a fusion study of the afore-

mentioned cutting-edge techniques to enhance ESP anomaly monitoring performance is a highly significant attempt and investigation.

Furthermore, fixed thresholds (FT) are frequently used in traditional methods to identify anomalies. However, the FT is not appropriate for ESP abnormal detection tasks since ESP has been operating in complex well conditions for a considerable amount of time, and the data is severely contaminated by erratic disturbance, which would result in a lot of false alarms and significantly impede the production schedule. Recent research has presented several adaptive threshold generation (ATG) approaches to address this problem [23], [30], [31]. In [30], adaptive thresholds are constructed by combining an empirical constant with support vector regression (SVR). The empirical constant and kernel function employed in SVR, however, have a significant impact on its efficiency. In [31], the optimal relevance vector machine's (RVM) probabilistic output is used to construct adaptive thresholds, which enable the wind turbine pitch system's status monitoring to function well. Due to the probabilistic output's reliance on the normal distribution assumption, which ESP might not meet, the application of this method for ESP anomaly identification is regrettably restricted. More recently, an XGBoost-regressor-based adaptive threshold adjustment (ATA) mechanism has been developed for ESP anomaly detection [23]. However, it is important to note that the XGBoost regressor is fundamentally a boosting algorithm, which tends to be more sensitive to noise and prone to overfitting. Additionally, it encompasses a complex hyperparameter system that necessitates meticulous tuning to achieve optimal performance. Thus, how to design an ATG method that minimizes sensitivity to noise while facilitating easier hyperparameter adjustments becomes a key challenge for ESP abnormality detection.

In this article, we present a novel approach for ESP anomaly detection, called dual spatio-temporal contrastive learning network with ATG (DSTCL-ATG), to address the aforementioned issues. The main contributions of this article are summarized below.

- 1) An integrated algorithm named DSTCL network is developed, which can effectively accomplish ESP normal process modeling with an excellent performance by introducing Crossformer's encoder-based spatio-temporal feature extraction into dual-path contrastive learning (DCL) architecture.
- 2) A new random-forest-regressor-based ATG method is investigated and used to improve frequent false alarms caused by erratic disturbance in the ESP system.
- 3) Practical data collected from four different ESP wells in a real oilfield site is used to evaluate the effectiveness and superiority of the proposed method. The experimental results demonstrate that the proposed method outperforms other state-of-the-art methodologies.

The rest of this article is structured as follows. The ESP system is briefly introduced in Section II. Section III elaborates on the proposed method. In Section IV, the effectiveness of

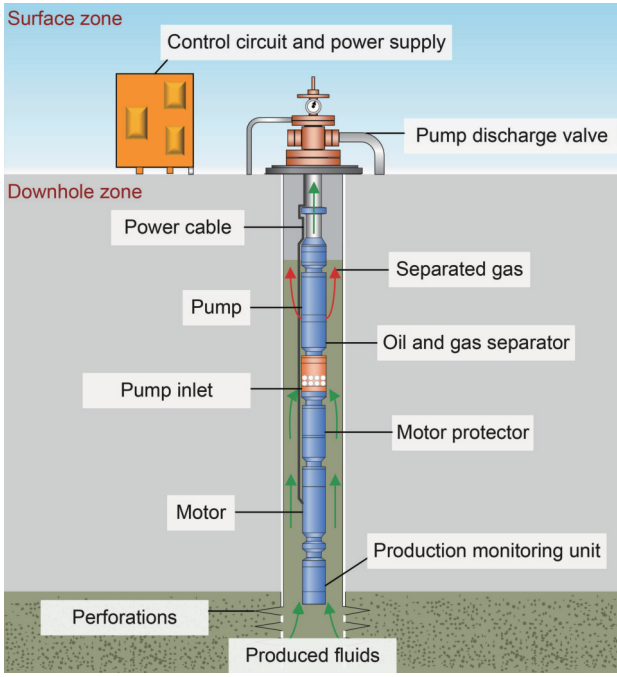


Fig. 1. Schematic representation of an ESP system [2].

the proposed method is validated. Finally, the conclusion is given in Section V.

II. BRIEF INTRODUCTION OF ESP SYSTEM

The ESP system is made up of two sections, the surface zone and the downhole zone, as seen in Fig. 1 [2]. Electric drives and pump control components, such as the power supply, control circuit, and pump discharge valve, are the primary components of the ESP system's surface zone. These components sense variations in flow, frequency, current, and voltage. The production monitoring device, motor, oil and gas separator, pump, and motor protector are all part of the ESP system's downhole zone. The motor is the central component of the system, serving as its power source. The motor protector keeps the motor separate from the fluid in the well. Crude oil enters the oil and gas separator through the pump input during the oil production process to accomplish gas-liquid separation. The separated gas enters the annulus when the oil is pumped to the surface zone.

ESP system operational status can be monitored with the help of the downhole production monitoring system, which currently has the ability to gather real-time process data, such as pressure, temperature, vibration, pump inlet, and outlet pressure differential. However, nonlinear responses in pump performance may result from changes in fluid viscosity, density, and gas-to-liquid ratio [32]. Additionally, long-term dynamic behavior is observed as the system responds to changes in fluid composition, reservoir pressure, temperature, operating conditions, and external factors [33]. Uncertain disturbances can also cause non-stationary fluctuations in normal operating conditions. The multivariate monitoring data naturally exhibits these complex features, which make ESP anomaly detection extremely difficult.

III. METHODOLOGY

A. Overview of the Proposed Method

To improve ESP anomaly detection performance, it is necessary to consider the characteristics of multiple variables, nonlinearity, long-term dynamics, and uncertain disturbances of the ESP system. In this section, we develop a contrastive learning-based technique for ESP anomaly detection. The proposed DSTCL-ATG approach consists of two modules: DSTCL and ATG, as illustrated in Fig. 2. To fully account for the multivariate, nonlinear, and long-term dependency characteristics of ESP data and achieve efficient unsupervised modeling of ESP normal operating conditions, the DSTCL module builds on a dual-branch contrastive learning architecture and introduces an encoder network of Crossformer to extract ESP data's spatio-temporal coupling features. Furthermore, instead of the more usual FT, the ATG module generates adaptive thresholds using a random forest regressor to detect whether ESP operating conditions are abnormal. This might reduce frequent false alarms caused by uncertain disturbances. The joint use of two modules is intended to enhance ESP anomaly detection capability.

B. Dual Spatio-Temporal Contrastive Learning

Most contrastive learning methods concentrate on acquiring feature representations by distinguishing between pairs of samples [34]. This characteristic renders them particularly well-suited for downstream tasks such as time-series classification [35]. However, the normal process modeling for ESP anomaly detection is fundamentally characterized as time-series forecasting. The primary objective here is to predict future operating conditions based on historical time windows rather than merely differentiating between instances. Therefore, the features learned through instance discrimination may be inadequate for achieving accurate forecasting. Recently, SimTS, a simple representation learning framework for time-series forecasting, has been proposed [36]. This framework employs a convolutional encoder-based DCL architecture to learn representations and performs exceptionally well in time-series prediction. However, beyond temporal correlations, variables with the same physical characteristics and spatially adjacent variables with distinct characteristics are also associated. This indicates that ESP data exhibits a complex spatio-temporal coupling. When applying the SimTS approach to describe ESP normal processes, the ESP data's peculiarities are not adequately considered. Fortunately, Crossformer utilizes a dual-layer attention mechanism to effectively capture spatio-temporal coupling within time-series data [26]. Consequently, we create a DSTCL module that incorporates the Crossformer encoder network into the DCL architecture to better capture spatio-temporal correlations in ESP data.

The developed DSTCL module comprises two branches: one dedicated to future forecasting and the other focused on future encoding. The future forecasting branch aims to map historical ESP time-series segments into predicted future representations, whereas the future encoding branch serves to encode forthcoming time-series segments to retrieve the actual future representation. Unsupervised ESP process modeling can

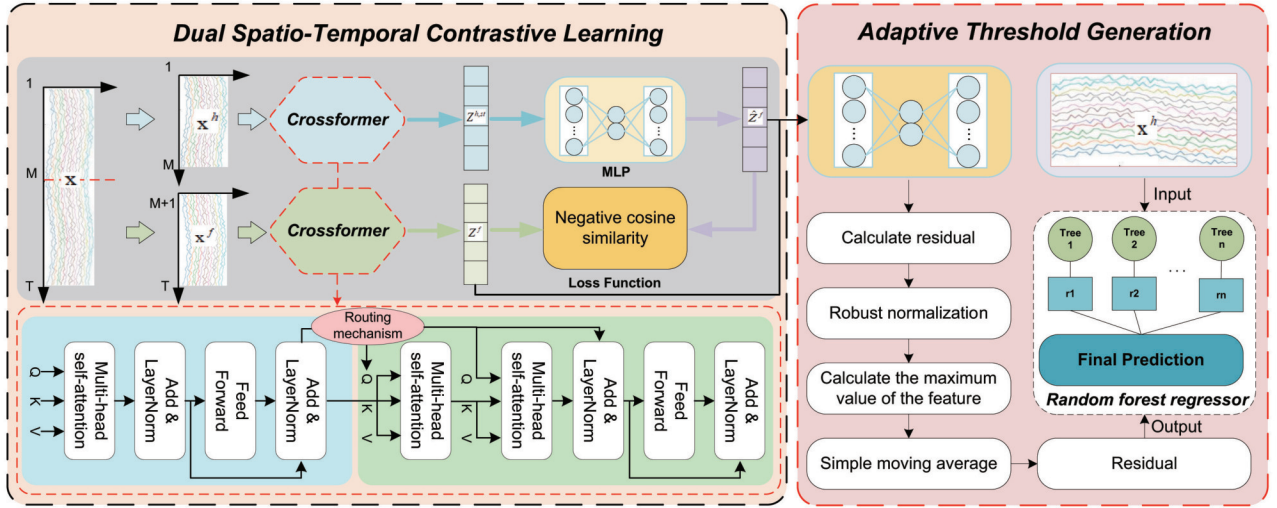


Fig. 2. Framework of the proposed DSTCL-ATG method.

be accomplished well by minimizing the residuals between two distinct future representations. This lays the groundwork for identifying ESP anomalies based on residual-adaptive threshold disparities.

To be more specific, given a raw ESP time-series data $\mathbf{x} = [x_1, x_2, \dots, x_T] \in \mathbb{R}^{C \times T}$, where C represents the number of variables and T denotes the length of the time series. The ESP data $\mathbf{x}$ can be partitioned into two components: historical segments $\mathbf{x}^h = [x_1, x_2, \dots, x_M] \in \mathbb{R}^{C \times M}$ and future segments $\mathbf{x}^f = [x_{M+1}, x_{M+2}, \dots, x_T] \in \mathbb{R}^{C \times (T-M)}$, with $0 < M < T$. In the future forecasting branch, the historical segment $\mathbf{x}^h$ will be fed into an encoder network G_ϕ to learn the historical latent spatio-temporal feature representations $\mathbf{Z}^{h, st} \in \mathbb{R}^{C \times M}$

$$\mathbf{Z}^{h, st} = G_\phi(\mathbf{x}^h) \quad (1)$$

where ϕ are the learnable parameters of the encoder network G_ϕ .

In practice, the core architecture of Crossformer is employed here as the encoder network G_ϕ . The operation details are presented as follows. $\mathbf{x}^h$ is first divided into L subsequences, represented as $\mathbf{x}^h \in \mathbb{R}^{C \times L \times (M/L)}$. Then, a multilayer perceptron (MLP) network is used to map them into the embedding sequences $\mathbf{X}^h \in \mathbb{R}^{C \times L \times Td}$

$$\mathbf{X}^h = \text{MLP}(\mathbf{x}^h) \quad (2)$$

where Td represents the dimension of the embedded sequences along the timeline. However, the above embedding sequences do not mark the sequential order relation of the subsequences; therefore, positional embedding $\mathbf{E}^h \in \mathbb{R}^{C \times L \times Td}$ is added to $\mathbf{X}^h$ to generate the tokenized embedding sequences $\mathbf{Z}^h \in \mathbb{R}^{C \times L \times Td}$

$$\mathbf{Z}^h = \mathbf{X}^h + \mathbf{E}^h. \quad (3)$$

To facilitate subsequent operations, the first two dimensions of $\mathbf{Z}^h$ are swapped. $\mathbf{Z}^h$ will be rewritten as $\mathbf{Z}^h \in \mathbb{R}^{L \times C \times Td}$, which will serve as the input of cross-time and cross-variable attention layers in the encoder network of Crossformer. More detailed, we define $\mathbf{Z}_{l, :}^h$ as the vectors of all variables at the l th subsequence, and $\mathbf{Z}_{:, c}^h$ for those of all subsequences in variable

c . Each variable will be fed into the multihead self-attention (MSA) [24] layer with Hd heads in the cross-time stage

$$\hat{\mathbf{Z}}_{:, c}^{h, time} = \text{LN} \left\{ \mathbf{Z}_{:, c}^h + \text{MSA}^{time}(\mathbf{Z}_{:, c}^h, \mathbf{Z}_{:, c}^h, \mathbf{Z}_{:, c}^h) \right\} \quad (4)$$

$$\mathbf{Z}^{h, time} = \text{LN} \left\{ \hat{\mathbf{Z}}^{h, time} + \text{MLP}(\hat{\mathbf{Z}}^{h, time}) \right\} \quad (5)$$

where $1 \leq c \leq C$ and LN indicate layer normalization [37]. The MSA layer is denoted by $\text{MSA}(Q, K, V)$, where Q, K , and V represent queries, keys, and values. Every variable has a single MSA layer. Both the MSA and MLP outputs are shown by $\hat{\mathbf{Z}}^{h, time}$ and $\mathbf{Z}^{h, time}$.

Following this step, $\mathbf{Z}^{h, time}$ records the dependencies between time-series segments in the same variable. The cross-variable stage then uses $\mathbf{Z}^{h, time}$ as its input to capture cross-variable dependency. In particular, several learnable routers initially aggregate messages from all variables utilizing routers as queries in MSA and all-variable vectors as keys and values. After that, the received messages are distributed among variables by routers using variable vectors as queries and aggregated messages as keys and values. By doing so, the all-to-all interactions among C variables are built

$$\mathbf{B}_{l, :}^h = \text{MSA}_1^{var}(\mathbf{R}_{l, :}^h, \mathbf{Z}_{l, :}^{h, time}, \mathbf{Z}_{l, :}^{h, time}), \quad 1 \leq l \leq L \quad (6)$$

$$\bar{\mathbf{Z}}_{l, :}^{h, var} = \text{MSA}_2^{var}(\mathbf{Z}_{l, :}^{h, time}, \mathbf{B}_{l, :}^h, \mathbf{B}_{l, :}^h), \quad 1 \leq l \leq L \quad (7)$$

$$\hat{\mathbf{Z}}^{h, st} = \text{LN}(\mathbf{Z}^{h, time} + \bar{\mathbf{Z}}^{h, var}) \quad (8)$$

$$\mathbf{Z}^{h, st} = \text{LN}(\hat{\mathbf{Z}}^{h, st} + \text{MLP}(\hat{\mathbf{Z}}^{h, st})) \quad (9)$$

where $\mathbf{R}^h \in \mathbb{R}^{L \times \text{cons} \times Td}$ ($\text{cons} \ll C$ is a small constant) is the trainable router matrix that dynamically learns to aggregate cross-variable information. During the cross-variable attention stage, this matrix first collects global features from all variables, as shown in (6), then redistributes them through attention weights, as shown in (7). $\mathbf{B}^h \in \mathbb{R}^{L \times \text{cons} \times Td}$ represents the aggregated messages from all variables. $\bar{\mathbf{Z}}^{h, var}$ represents the router's output. $\hat{\mathbf{Z}}^{h, st}$ and $\mathbf{Z}^{h, st}$ signify the output of residual connection and MLP, respectively.

By repeating the above cross-time and cross-variable stages α times, executing necessary dimension swapping, and reasonably selecting MLP parameters, the spatio-temporal feature representation of the historical segments $\mathbf{Z}^{h, \text{st}} \in \mathbb{R}^{C \times M}$ can be obtained. After that, a predicted MLP network F_θ is applied to $\mathbf{Z}^{h, \text{st}}$ to predict the future representation $\hat{\mathbf{Z}}^f \in \mathbb{R}^{C \times (T-M)}$

$$\hat{\mathbf{Z}}^f = F_\theta(\mathbf{Z}^{h, \text{st}}) \quad (10)$$

where θ denotes the learnable parameters of the MLP network F_θ .

Similarly, in the future encoding branch, the future segments $\mathbf{x}^f$ are directly encoded by the encoder network of Crossformer G_ϕ to generate the actual future representation $\mathbf{Z}^f \in \mathbb{R}^{C \times (T-M)}$

$$\mathbf{Z}^f = G_\phi(\mathbf{x}^f). \quad (11)$$

Note that to guarantee that the dimension is appropriate, M is set to $1/2T$ in our setup.

After obtaining the predicted future representation $\hat{\mathbf{Z}}^f$ and the actual future representation $\mathbf{Z}^f$, cosine distance $d_{\cos}$ is employed to evaluate the similarity between these two representations. Based on this metric, a loss function $\mathcal{L}$ is formulated to facilitate the ESP spatio-temporal feature representation learning

$$\begin{aligned} \mathcal{L}(\phi, \theta) &= -d_{\cos}(\hat{\mathbf{Z}}^f, \mathbf{Z}^f) \\ &= -\frac{1}{T-M} \sum_{m=M+1}^T \frac{\hat{\mathbf{Z}}_{c,m}^f \cdot \mathbf{Z}_{c,m}^f}{\|\hat{\mathbf{Z}}_{c,m}^f\| \|\mathbf{Z}_{c,m}^f\|} \end{aligned} \quad (12)$$

where $\hat{\mathbf{Z}}_{c,m}^f$ and $\mathbf{Z}_{c,m}^f$ denote the predicted future representation and the actual future representation of the m th time instant in the c spatial dimension, respectively. The notation $\|\cdot\|$ signifies the $L2$ norm.

Considering that the same encoder network is employed in the real optimization process, model optimization may inadvertently steer the actual future representation $\mathbf{Z}^f$ toward the predicted future representation $\hat{\mathbf{Z}}^f$, which is undesirable. To mitigate this issue, a stop-gradient approach [38] is implemented in the future encoding branch. This strategy ensures that the encoder network does not receive updates from the actual future representation but only anticipates features based on historical data.

After training the DSTCL module, the historical latent spatio-temporal feature representations of ESP $\mathbf{Z}^{h, \text{st}}$ will be obtained. An MLP network-based single-point predictor F_γ will be designed, where the feature representation following the encoded history segment $\mathbf{Z}^{h, \text{st}}$ is utilized to predict the ESP's single-point operational state after this history segment $\hat{x}_{M+1}$:

$$\hat{x}_{M+1} = F_\gamma(\mathbf{Z}^{h, \text{st}}) \quad (13)$$

where γ denotes the learnable parameters of the predictor F_γ .

Finally, the mean square error (mse) between the predicted single-point operating state $\hat{x}_m$ and its actual operating state x_m serves as the loss function for optimizing the single-point predictor

$$\mathcal{L}(\gamma) = -\frac{1}{T-M} \sum_{m=M+1}^T \|\hat{x}_m - x_m\|_2^2. \quad (14)$$

When the model parameters have been learned, the predictor output $\hat{x}_m$ w.r.t. $m \in \{M+1, M+2, \dots, T\}$ can be estimated based on the past M observational data.

Remark 1: Notably, the primary contribution of DSTCL lies in the integration of the encoder network from Crossformer into a DCL architecture. This approach facilitates the capture of spatio-temporal dependencies inherent in ESP data and allows for the modeling of normal ESP processes within an unsupervised contrastive learning paradigm, which might be conducive to ESP anomaly detection. In addition, two key distinctions exist between DSTCL and the previously developed TAN framework [23]. First, while TAN solely employs the attention mechanism along the temporal dimension, DSTCL utilizes Crossformer to implement dual attention mechanisms across both temporal and variable dimensions. Second, whereas TAN adopts an AE reconstruction architecture, DSTCL combines unsupervised contrastive learning with a single-point prediction architecture. This integrated approach enables more effective acquisition of invariant features in ESP processes, thereby achieving superior process modeling performance.

C. Adaptive Threshold Generation

The proposed DSTCL module can be used to model the complex ESP process, but it inevitably learns some uncertain disturbances in the modeling process, which will affect the model's output. In this case, using residuals exceeding the predetermined threshold for alarms will result in many false alarms, as the threshold cannot adaptively change in the case of uncertain interference, causing significant fluctuations in the state of the ESP. In addition, setting the threshold too high can easily miss anomalies, which is very dangerous. Therefore, it is crucial to create an ATG method to counteract the negative effects of interference and improve the ESP anomaly detection performance.

In general, the threshold is determined by taking the maximum residual value from the validation set as a fixed reference point. During the testing phase with sample inputs, it is anticipated that the model will be capable of distinguishing anomalies by employing varying thresholds tailored to different inputs. This necessitates that the model generate distinct residuals based on these diverse inputs in real time. An intuitive approach to achieving this goal involves establishing a regressor mapping between the inputs and their corresponding residuals. In our ATG module, we utilize a random forest regressor to build this relationship and train the residual generator using data from the validation set.

Specifically, the residual r_c^{val} between the predicted state for each spatial feature dimension within the validation set $\hat{x}_{c,m}^{\text{val}}$ and its corresponding ground truth $x_{c,m}^{\text{val}}$ is first calculated by

$$r_c^{\text{val}} = |\hat{x}_{c,m}^{\text{val}} - x_{c,m}^{\text{val}}| \quad (15)$$

where $\hat{x}_{c,m}^{\text{val}}$, $x_{c,m}^{\text{val}}$, and r_c^{val} have been appended with the "val" superscripts to clarify that the computations are derived from the validation set. To address the scale discrepancies of

residuals across various spatial features, a robust normalization technique [39] is employed

$$\tilde{r}_c^{\text{val}} = \frac{r_c^{\text{val}} - p_c^{\text{val}}}{q_c^{\text{val}}} \quad (16)$$

where p_c^{val} and q_c^{val} are the median and inter-quartile range of r_c^{val} , respectively.

Then, the maximum value of $\tilde{r}_c^{\text{val}}$ across all spatial feature dimensions is computed

$$r^{\text{val}} = \max_c \tilde{r}_c^{\text{val}}. \quad (17)$$

To enhance the robustness of the normalized residual against disturbances under normal conditions, a simple moving average (SMA) approach [40] is used to generate the smoothed residual r_s^{val}

$$r_s^{\text{val}} = \text{SMA}(r^{\text{val}}). \quad (18)$$

Ultimately, the random forest regressor F_ψ is trained to create mappings from $\mathbf{x}^{h,\text{val}}$ to r_s^{val}

$$r_s^{\text{val}} = F_\psi(\mathbf{x}^{h,\text{val}}) \quad (19)$$

where $\mathbf{x}^{h,\text{val}} = [x_1^{\text{val}}, x_2^{\text{val}}, \dots, x_M^{\text{val}}] \in \mathbb{R}^{C \times M}$ represents the historical segments in the validation set that serve as input to F_ψ , r_s^{val} denotes the smoothed residual in the validation set that serves as output to F_ψ , and ψ refers to the model parameters. The number of decision trees in random forest nt and the depth of the decision tree dt are selected according to the performance of anomaly detection. The optimally trained random forest regressor F_ψ^* will be derived by minimizing the mse between the predicted outputs and the ground truth, which will subsequently be employed to generate adaptive thresholds during the online detection phase.

Remark 2: It should be noted that, in contrast to the XGBoost regressor-based ATA mechanism [23], which sequentially builds learners with increasing focus on previously misclassified (potentially noisy) samples, the proposed random forest regressor-based ATG module employs a bagging strategy that trains multiple decision trees independently on bootstrap samples. This ensemble approach inherently reduces variance by averaging predictions across diverse learners, thereby mitigating the influence of outliers and noisy data. Furthermore, the feature randomness introduced at each split decorrelates the trees, enhancing the model's robustness to noise. These theoretical characteristics collectively make random forest a more noise-tolerant choice for adaptive threshold estimation in ESP anomaly detection. In addition, it requires adjustment of only a few key parameters (e.g., number of decision trees and maximum depth), while demonstrating smoother performance variation across hyperparameter configurations. This characteristic may significantly enhance ESP's anomaly detection capabilities.

D. DSTCL-ATG-Based ESP Anomaly Detection

The flowchart for ESP anomaly detection, based on the DSTCL-ATG methodology, is presented in Fig. 3. As indicated by the red and blue dotted boxes, the proposed monitoring framework is organized into two distinct phases: offline training and online detection.

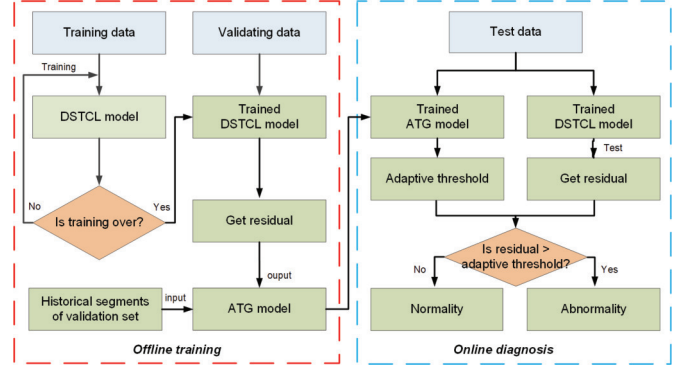


Fig. 3. Flowchart of DSTCL-ATG-based ESP anomaly detection.

In the offline training phase, the training data $\mathbf{x}^{\text{train}}$ is initially input into the designed DSTCL module to learn the optimal model parameters θ^* , ϕ^* and γ^* . Subsequently, historical segments of validation data $\mathbf{x}^{h,\text{val}}$ are processed through the well-trained DSTCL module to obtain the predicted states $\hat{\mathbf{x}}^{\text{val}}$. Following this, the smoothed residuals r_s^{val} will be computed using (15)–(18). Finally, sample pairs constructed from $\mathbf{x}^{h,\text{val}}$ and r_s^{val} will be utilized to train the random forest regressor-based ATG module until the optimal parameters ψ^* are achieved.

In the online detection stage, historical segments of test data $\mathbf{x}^{h,\text{test}}$ are first fed into the trained DSTCL module to obtain the predicted state $\hat{\mathbf{x}}^{\text{test}}$. Then, the smoothed residual r_s^{test} is computed using (15)–(18) in the test set. Afterward, the adaptive threshold $\mathcal{J}_{\text{th}}$ is generated according to (19), and the empirical statistical principle that nearly 99.73% of data in a Gaussian distribution lies within the $\mu \pm 3\sigma$ range. Motivated by this, we incorporate a $3\sigma_{\text{val}}$ margin to enhance disturbance mitigation capability

$$\mathcal{J}_{\text{th}} = F_{\psi^*}(\mathbf{x}^{h,\text{test}}) + 3\sigma_{\text{val}} \quad (20)$$

where σ_{val} represents the standard deviation of the residual of the validation set.

Finally, we will assess whether the ESP process exhibits any abnormalities in accordance with the following criteria:

$$\begin{cases} r_s^{\text{test}} > \mathcal{J}_{\text{th}} & \Rightarrow \text{abnormality} \\ r_s^{\text{test}} \leq \mathcal{J}_{\text{th}} & \Rightarrow \text{normality.} \end{cases} \quad (21)$$

When an abnormal ESP operating condition is effectively monitored, the operator can implement appropriate measures to avoid potential losses. Detailed processes are summarized in Algorithms 1 and 2.

IV. EXPERIMENTS

A. Data Description

To assess the effectiveness of the proposed method, we collect ESP operational data from four distinct anomalous wells belonging to a real production company. These wells are designated as #B20H3, #B09ST1, #B18ST1, and #A03ST3, which correspond to specific faults: emulsification, electrical failure, plugging, and shaft breakage, respectively. Table I summarizes the specific time when the exception occurred,

TABLE I
EXPERIMENTAL DATA DESCRIPTION

ESP No.	Start time	Ending time	Abnormal log	Test data start time	Sampling interval	Variable number
#B20H3	00:00, Sep.01, 2022	00:00, Sep.15, 2022	04:20, Sep.13, 2022	00:00, Sep.10, 2022	5 s	15
#B09ST1	00:00, May.22, 2022	00:00, Jun.08, 2022	22:50, Jun.06, 2022	00:00, Jun.01, 2022	5 s	15
#B18ST1	00:00, Aug.20, 2021	00:00, Sep.06, 2021	20:00, Sep.03, 2021	00:00, Sep.01, 2021	5 s	15
#A03ST3	00:00, May.30, 2021	00:00, Jun.15, 2021	23:50, Jun.11, 2021	00:00, Jun.08, 2021	5 s	15

Algorithm 1 Learning Algorithm of DSTCL-ATG

Input:

A mini-batch $D_{train} = \{\mathbf{x}_i^{train}\}_{i \in [1:N]}$ with N time-series samples;

Initial model parameters ϕ , θ , γ , and ψ

Output: ϕ^* , θ^* , γ^* , and ψ^*

DSTCL model training

- 1: **for** number of training iterations **do**
- 2: **for** number of batch sizes **do**
- 3: **for** $\mathbf{x}^{train}$ in D_{train} **do**
- 4: $\mathbf{x}^h, \mathbf{x}^f = \mathbf{x}^{train}[:, :M, :], \mathbf{x}^{train}[:, M+1 :, :]$
- 5: Calculate $\mathbf{Z}^{h,st}$ and $\mathbf{Z}^f$ by (1) and (11)
- 6: Obtain $\hat{\mathbf{Z}}^f$ by (10) and stop gradient
- 7: Compute $\mathcal{L}(\phi, \theta)$ by (12)
- 8: **end for**
- 9: **end for**
- 10: Update parameters ϕ and θ
- 11: **end for**
- 12: Obtain the trained parameters ϕ^* and θ^*
- 13: Train single-point predictor F_γ by (13)–(14) and obtain the trained parameters γ^*
- 14: # *ATG model training*
- 15: Calculate validation set's residual r_s^{val} by (15)–(18)
- 16: Learn the random forest regressor F_ψ by (19)
- 17: Obtain the trained parameters ψ^*
- 18: **return** ϕ^* , θ^* , γ^* , and ψ^*

Algorithm 2 Anomaly Detection Based on DSTCL-ATG

Input:

Testing data $\mathbf{x}^{h,test}$

Trained parameters ϕ^* , θ^* , γ^* , and ψ^*

Output: Detection results

- 1: Calculate test set's residual r_s^{test} by (15)–(18)
- 2: Generate adaptive threshold $\mathcal{J}_{th}$ by (20)
- 3: Obtain detection results by (21);
- 4: **return** detection results

the data sampling time, the monitoring number of variables contained in each piece of data, etc. More details of the monitoring variables are illustrated in Table II, where variables 1-7 correspond to the production attributes of the oil well, whereas variables 8-15 pertain to the operational characteristics of the equipment. It is noteworthy that in all datasets used in our research, there is no overlap between the training and test sets. To enhance the model's detection effectiveness and data quality, raw data is scaled using the standard normalization criterion.

TABLE II
MONITORING VARIABLE DESCRIPTION

No.	Variable	Unit
#1	ESP intake pressure	MPa
#2	Oil pressure	MPa
#3	ESP discharge pressure	MPa
#4	Casing pressure	MPa
#5	Nozzle diameter	Mm
#6	Wellhead temperature	°C
#7	ESP intake temperature	°C
#8	Voltage	V
#9	Leakage current	mA
#10	Motor power	kw
#11	Motor temperature	°C
#12	Pump torque current	A
#13	Vibration	G
#14	Current	A
#15	Frequency	Hz

B. Experimental Setup

1) *Baseline Methods*: In our experiments, we compare the proposed method with several state-of-the-art abnormality detection techniques, including PCA [16], DEEP-IFOREST [41], MEMTO [42], NPSR [43], TAN-ATA [23], and STEN [44]. More details of comparative methods are presented below.

- 1) *PCA* [16]: PCA facilitates the projection of high-dimensional ESP monitoring data into a low-dimensional subspace by maximizing the variance of the predictive dimensions. Ultimately, anomalies are detected by assessing whether the statistical properties of features in this low-dimensional space exceed predefined limits.
- 2) *Deep isolation forest (DEEP-IFOREST)* [41]: To accomplish anomaly isolation, DEEP-IFOREST employs random axis-parallel cutting after mapping the original data into a random representation set using NNs and utilizes deviation-enhanced anomaly scoring to recognize anomalies.
- 3) *Memory-guided transformer (MEMTO)* [42]: MEMTO is a reconstruction-based approach, which incorporates a gated memory module, a two-stage training paradigm, and a bi-dimensional deviation-based detection criterion to achieve time-series anomaly detection.
- 4) *TAN-ATA* [23]: TAN with ATA integrates a temporal-attention network built upon an LSTM-AE architecture with an XGBoost-regressor-based ATA mechanism, delivering enhanced ESP anomaly detection through this hybrid strategy.

TABLE III
COMPARISON RESULTS OF DIFFERENT METHODS ON FOUR ESP WELL DATASETS

Method	#B20H3			#B09ST1			#B18ST1			#A03ST3		
	<i>FAR</i>	<i>FDR</i>	<i>g_{mean}</i>	<i>FAR</i>	<i>FDR</i>	<i>g_{mean}</i>	<i>FAR</i>	<i>FDR</i>	<i>g_{mean}</i>	<i>FAR</i>	<i>FDR</i>	<i>g_{mean}</i>
PCA [16]	0.0003	0.8537	0.9238	0.0685	0.9179	0.9247	0.0559	0.9157	0.9298	0.0093	0.8513	0.9184
DEEP-IFOREST [41]	0.0150	0.9145	0.9491	0.0415	0.9648	0.9616	0.0285	0.9165	0.9436	0.0201	0.9448	0.9622
MEMTO [42]	0.0719	0.9932	0.9601	0.0023	0.9810	0.9893	0.0311	0.9581	0.9634	0.0033	0.9418	0.9689
NPSR [43]	0.0151	0.9504	0.9675	0.0093	0.9905	0.9906	0.0376	0.9769	0.9696	0.0182	0.9998	0.9906
TAN-ATA [23]	0.0404	0.9614	0.9605	0.0101	0.9907	0.9903	0.0323	0.9724	0.9701	0.0014	0.9851	0.9918
STEN [44]	0.0380	0.9626	0.9623	0.0171	1.0000	0.9914	0.0362	0.9774	0.9706	0.0033	0.9908	0.9937
DSTCL-ATG	0.0350	0.9935	0.9791	0.0024	1.0000	0.9988	0.0257	1.0000	0.9871	0.0012	1.0000	0.9994

- 5) *NPSR* [43]: Nominality score conditioned time-series anomaly detection by point/sequential reconstruction (NPSR) is an unsupervised approach that utilizes point-based and sequence-based reconstruction models, which outperforms most representative baselines for time-series anomaly detection.
- 6) *Spatial-temporal normality learning (STEN)* [44]: STEN comprises two distinct modules designed to capture the inherent temporal and spatial correlations present within sequences. By integrating these two modules, STEN facilitates effective modeling of normal processes, thereby enhancing the performance of anomaly detection.

To be more specific, hyperparameters of the baseline methods are configured as follows. PCA uses three principal components. DEEP-IFOREST sets the expected proportion of anomalies to 0.01. MEMTO employs 32 memory items with an embedding dimension of 64 and initializes without memory item embedding. TAN-ATA implements 15 LSTM units in the output layer, a hidden layer dimension of 64 500 trees in XGBoost, and a maximum tree depth of 10. NPSR uses a hidden dimension of 64 with one encoder layer. STEN adopts a temporal hidden dimension of 64, a Transformer spatial encoder, a mask ratio of 0.1, and one temporal layer.

2) *Evaluation Indicator*: In our experiments, the performance of anomaly detection is comprehensively assessed using three metrics: false alarm rate (FAR), fault detection rate (FDR), and their geometric mean (g_{mean}), as advocated by [45]. The formulations for these three indicators are presented below

$$\text{FAR} = N_n^f / N_n \quad (22)$$

$$\text{FDR} = N_f^d / N_f \quad (23)$$

$$g_{\text{mean}} = \sqrt{\text{FDR} \times (1 - \text{FAR})} \quad (24)$$

where N_f^d denotes the number of detected abnormal samples, while N_n^f indicates the number of normal samples identified through false alarms. Additionally, N_f and N_n represent the total counts of abnormal and normal samples, respectively.

3) *Implementation Details*: In our studies, we employ the sliding window approach to acquire ESP time-series segments, with the window size T set to 300 and the M set to 150. In the DSTCL module, based on the findings from multiple experiments, the dimensions of the embedded sequences throughout

the timeline Td and the number of Crossformer encoder blocks α are set to 64 and 1, respectively. In the Crossformer encoder block, L , Hd , and cons are set to 5, 4, and 10, respectively. The model is trained via a stochastic gradient descent (SGD) optimizer with a learning rate of $1e-3$ and a weight decay of $1e-4$. All experiments are conducted throughout 100 epochs. In the ATG module, the number of decision trees nt and the depth of the decision tree dt are set to 100 and 10, respectively. All experiments in this work are implemented based on Python 3.7 with PyTorch 1.8 and performed on Windows 11 with an Intel¹ Core² GeForce RTX3060 CPU at 2.30 GHz with 16 GB of RAM.

C. Comparative Analysis

The comparative experimental results of various methods applied to data from four different wells are summarized in Table III. Generally, a smaller FAR is associated with a larger FDR, and a higher g_{mean} indicates better anomaly detection performance. From the table, it is evident that the Deep IFOREST, a machine learning-based method, outperforms the PCA-based multivariate statistical analysis method in ESP anomaly detection. Additionally, the NN-based MEMTO, NPSR, and TAN-ATA methods surpass both PCA and Deep IForest in ESP anomaly monitoring performance, demonstrating that incorporating the dynamic characteristics of ESPs enhances detection performance. TAN-ATA, unlike MEMTO and NPSR, employs an adaptive threshold adjustment based on an XGBoost regressor to improve detection performance. This illustrates that the adaptive threshold method is quite successful. The STEN method, which takes into account both the spatial and temporal characteristics of the data, yields even better results. Notably, the proposed DSTCL-ATG method delivers superior performance compared to other advanced methods, confirming the effectiveness and advantages of the designed DSTCL and random-forest-regressor-based ATG techniques.

D. Effects of the DSTCL

Table IV presents the comparative results of the average residuals between the DSTCL model and the DCL model on the datasets from four ESP wells. As shown in the table,

¹Registered trademark.

²Trademarked.

TABLE IV
COMPARISON RESULTS OF AVERAGE RESIDUALS WITH AND WITHOUT THE CROSSFORMER IN DSTCL ON FOUR ESP WELL DATASETS

Variable	#B20H3		#B09ST1		#B18ST1		#A03ST3	
	<i>DCL</i>	<i>DSTCL</i>	<i>DCL</i>	<i>DSTCL</i>	<i>DCL</i>	<i>DSTCL</i>	<i>DCL</i>	<i>DSTCL</i>
1	0.9769	0.6947	0.7586	0.3952	0.7585	0.3959	0.7586	0.3956
2	0.6864	0.2893	0.9683	0.7489	0.9799	0.5773	0.9475	0.7924
3	0.8726	1.4063	1.0852	1.3131	1.0685	1.2938	1.0192	1.3715
4	0.9005	0.9228	0.9926	0.9625	1.0918	0.9915	1.1816	0.8899
5	0.9351	0.5856	1.0263	0.7707	0.9980	0.6255	0.7751	0.4371
6	1.0846	0.8992	1.1048	0.8873	0.9322	0.8458	1.0818	0.9111
7	1.0772	0.6816	1.0663	1.0057	1.2127	0.5941	0.9534	0.7129
8	0.8279	0.3353	1.0781	0.4532	1.0654	0.6039	1.0707	0.9481
9	1.0985	0.7557	1.0487	0.7959	0.7975	0.3481	0.7975	0.3480
10	1.0923	0.8985	1.1335	0.7576	1.0566	0.7890	0.8608	0.6411
11	1.0644	1.0103	0.9988	0.9995	1.1239	1.0433	0.8752	0.9497
12	1.0004	0.7823	0.7532	0.3070	1.0471	0.5629	0.7535	0.3073
13	1.1722	0.6154	1.0402	0.4116	0.8602	0.3750	0.8602	0.3747
14	0.7732	1.3562	0.7733	1.4499	1.0857	1.2244	1.0628	1.2696
15	1.1104	0.5677	1.1295	0.6471	1.1426	0.5113	0.8841	0.1350

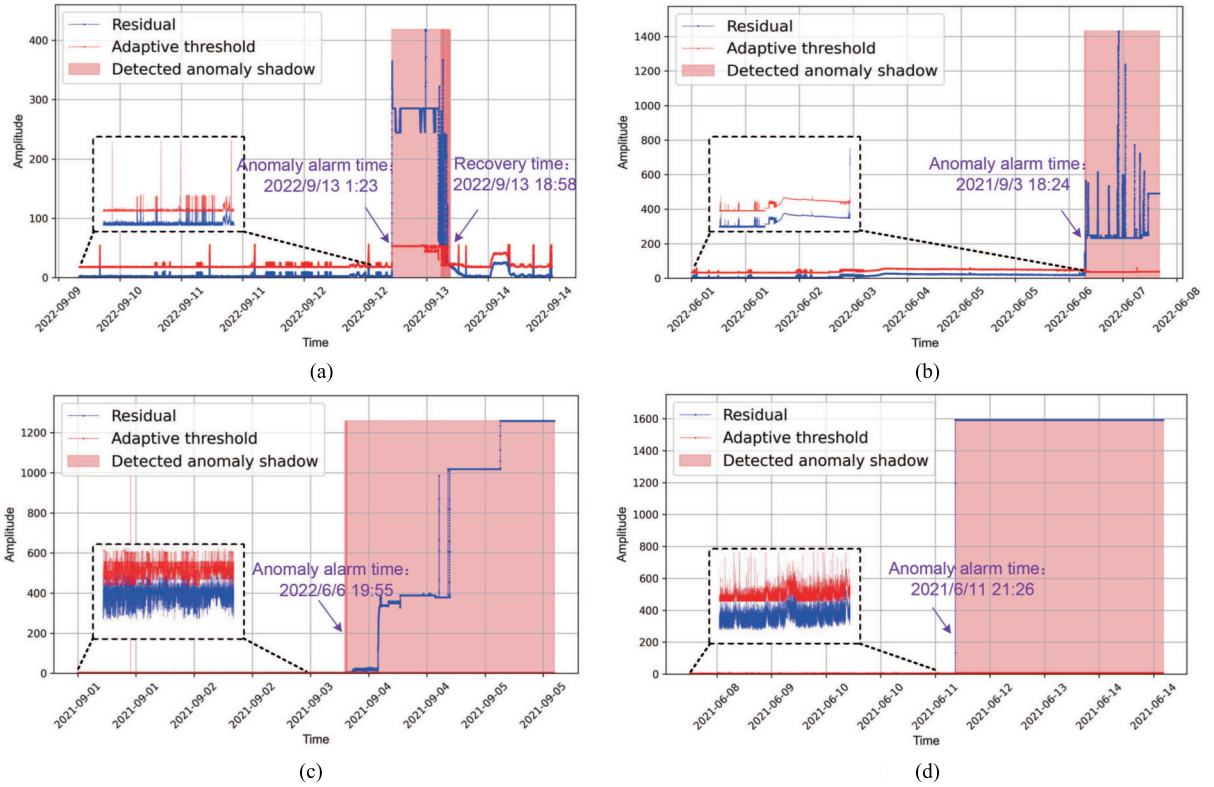


Fig. 4. Anomaly detection performance of the proposed DSTCL-ATG method on the test set of four different datasets. (a) B20H3, (b) B09ST1, (c) B18ST1, and (d) A03ST3.

while the average residuals of a few variables are marginally higher for certain wells, the residuals produced by the DSTCL model, enhanced by integrating the Crossformer architecture into the DCL framework, are consistently smaller than those of the DCL model. These results indicate that the proposed DSTCL model demonstrates superior accuracy in modeling the normal operational processes of ESP systems. This comparison effectively highlights the enhanced performance and robustness of the DSTCL model, underscoring its effectiveness and superiority.

E. Effects of the ATG

The anomaly detection performance of the proposed DSTCL-ATG method is intuitively illustrated in Fig. 4. As depicted in Fig. 4, when the ESP experiences an anomaly, the residuals exceed the adaptive threshold, triggering an alarm. The adaptive threshold is highly effective in mitigating false alarms caused by uncertain disturbances during the normal operation of the ESP. For instance, in the case of well #B20H3, the DSTCL-ATG method detected an anomaly at 1:23 AM on

TABLE V
ABNORMALITY DETECTION RESULTS UNDER DIFFERENT THRESHOLD SETTINGS

Method	#B20H3			#B09ST1			#B18ST1			#A03ST3		
	<i>FAR</i>	<i>FDR</i>	<i>gmean</i>	<i>FAR</i>	<i>FDR</i>	<i>gmean</i>	<i>FAR</i>	<i>FDR</i>	<i>gmean</i>	<i>FAR</i>	<i>FDR</i>	<i>gmean</i>
DSTCL-FT	0.0728	0.9998	0.9628	0.0310	1.0000	0.9844	0.0342	1.0000	0.9828	0.0176	1.0000	0.9911
DSTCL-SVR	0.0525	0.9997	0.9732	0.0244	1.0000	0.9877	0.0292	1.0000	0.9853	0.0105	1.0000	0.9948
DSTCL-RVM	0.0487	0.9997	0.9752	0.0199	1.0000	0.9900	0.0283	1.0000	0.9857	0.0072	1.0000	0.9964
DSTCL-DGP	0.0507	0.9997	0.9742	0.0218	1.0000	0.9890	0.0288	1.0000	0.9855	0.0132	1.0000	0.9934
DSTCL-BRF	0.0447	0.9997	0.9772	0.0065	1.0000	0.9968	0.0264	1.0000	0.9867	0.0040	1.0000	0.9980
DSTCL-ATG	0.0350	0.9935	0.9791	0.0024	1.0000	0.9988	0.0257	1.0000	0.9871	0.0012	1.0000	0.9994

September 13, 2022, approximately 3 h prior to the recorded anomaly detection time in the field log at 4:20 AM on the same day. Similarly, for wells #B09ST1, #B18ST1, and #A03ST1, the anomaly warning times provided by the DSTCL-ATG method were approximately 3, 2, and 2.5 h earlier than those logged manually, respectively. These findings demonstrate the ability of the proposed method to issue anomaly warnings significantly earlier, enhancing operational safety and response efficiency.

Moreover, we conduct comparative studies on different threshold-setting strategies, as shown in Table V. Although methods such as FT, SVR, and RVM achieve relatively high FDR, they result in significantly higher FAR, indicating a tendency to raise more false alarms under fluctuating ESP conditions. In addition to these traditional approaches, more advanced uncertainty-based methods like deep Gaussian process (DGP) [46] and Bayesian random forest (BRF) [47] also reach high FDR, but their FAR remain elevated. This may be attributed to two key factors: the increased model complexity introduced by the hierarchical structure of DGP and the inference variance introduced by the posterior sampling mechanism in BRF. Both factors may increase sensitivity to input noise and elevate the risk of overfitting, consequently impacting the stability of predictions during practical deployment. In contrast, the proposed lightweight random-forest-regressor-based ATG module achieves consistently high FDR while maintaining the lowest FAR and the highest g_{mean} across all four ESP well datasets. It provides a better balance between detection performance and model simplicity. The use of a bagging-based ensemble and a relatively shallow structure contributes to more stable value estimation and reduced variance in threshold prediction, making ATG more suitable for deployment in complex ESP operational environments.

F. Ablation Study

The proposed DSTCL-ATG technique consists primarily of the DCL architecture, Crossformer encoder, and random-forest-regressor-based ATG strategy. The following ablation experiments were carried out to ensure that each component of DSTCL-ATG was effective. Here, Method 1 substitutes the Crossformer encoder with a multiscale CNN as the encoder network G_ϕ , without employing the ATG method. Method 2 utilizes only the Crossformer NN as the ESP feature extractor and does not incorporate either the dual-branch contrastive learning architecture or the ATG strategy. Method 3

TABLE VI
ABLATION EXPERIMENTAL RESULTS OF DSTCL-ATG

Method	Model Structure					<i>FAR</i>	<i>FDR</i>	<i>gmean</i>
	DCL	Crossformer	TAN	RF	XGBoost			
Method 1	✓	×	×	×	×	0.0544	0.9506	0.9481
Method 2	×	✓	×	×	×	0.0629	0.9458	0.9414
Method 3	✓	✓	×	×	×	0.0838	0.9997	0.9570
Method 4	×	×	✓	✓	×	0.0451	0.9841	0.9694
Method 5	✓	✓	×	×	✓	0.0372	0.9944	0.9785
DSTCL-ATG	✓	✓	×	✓	×	0.0350	0.9935	0.9791

exclusively implements the well-designed DSTCL module while omitting the ATG technique. Method 4 employs TAN as the backbone network to substitute the DSTCL module while integrating the random-forest-regressor-based ATG mechanism. Method 5 replaces the random forest regressor with the XGBoost regressor based on the proposed DSTCL-ATG. In contrast, the proposed DSTCL-ATG approach integrates DSTCL and random-forest-regressor-based ATG effectively.

The ablation experimental method was implemented on well #B20H3, and the results for the FAR, FDR, and g_{means} indicators are presented in Table VI. From the results, we can draw the following observations.

- 1) The performance of Method 3 surpasses that of Method 2, indicating that the Crossformer encoder is superior to the multiscale CNN network in capturing spatio-temporal correlations from ESP data.
- 2) The superiority of Method 3 over Method 2 highlights the advantages of the DCL architecture.
- 3) The proposed DSTCL-ATG demonstrates better performance than Method 3, revealing the effectiveness of the ATG strategy.
- 4) DSTCL-ATG is superior to Method 4, which confirms that the proposed DSTCL module has greater ESP process modeling capability than TAN.
- 5) The failure of Method 5 to DSTCL-ATG implies that the random-forest-regressor-based ATG performs better in ESP anomaly detection than the XGBoost-regressor-based ATA in TAN-ATA.

These findings suggest that all designed modules are effective in enhancing ESP anomaly detection performance.

Furthermore, we conduct an evaluation of the performance of ESP anomaly detection under various prediction (including five-step and ten-step forecasting) and reconstruction (including full-window reconstruction) strategies employed in the

TABLE VII
COMPARISON RESULTS OF DSTCL-ATG USING DIFFERENT PREDICTION OR RECONSTRUCTION STRATEGIES ON FOUR ESP WELL DATASETS

Method	#B20H3			#B09ST1			#B18ST1			#A03ST3		
	<i>FAR</i>	<i>FDR</i>	<i>gmean</i>	<i>FAR</i>	<i>FDR</i>	<i>gmean</i>	<i>FAR</i>	<i>FDR</i>	<i>gmean</i>	<i>FAR</i>	<i>FDR</i>	<i>gmean</i>
Multi-step (5 steps)	0.0383	0.9317	0.9465	0.0028	1.0000	0.9985	0.0257	0.9999	0.9870	0.0005	1.0000	0.9997
Multi-step (10 steps)	0.0332	0.9947	0.9806	0.0023	1.0000	0.9980	0.0243	1.0000	0.9877	0.0032	1.0000	0.9983
Full-window reconstruction	0.0358	0.9956	0.9797	0.0024	1.0000	0.9987	0.0245	1.0000	0.9876	0.0028	1.0000	0.9985
Single-point	0.0350	0.9935	0.9791	0.0024	1.0000	0.9988	0.0257	1.0000	0.9871	0.0012	1.0000	0.9994

TABLE VIII
TIME CONSUMPTION OF DSTCL-ATG USING DIFFERENT PREDICTION OR RECONSTRUCTION STRATEGIES ON FOUR ESP WELL DATASETS

Method	#B20H3		#B09ST1		#B18ST1		#A03ST3	
	<i>Train(s)</i>	<i>Test(ms)</i>	<i>Train(s)</i>	<i>Test(ms)</i>	<i>Train(s)</i>	<i>Test(ms)</i>	<i>Train(s)</i>	<i>Test(ms)</i>
Multi-step (5 steps)	7.69	0.36	7.92	0.39	7.70	0.35	7.72	0.19
Multi-step (10 steps)	7.95	0.35	8.08	0.41	7.65	0.34	8.43	0.19
Full-window reconstruction	8.27	0.35	8.29	0.39	7.68	0.34	7.98	0.20
Single-point	7.87	0.35	8.09	0.38	7.49	0.34	7.97	0.19

TABLE IX
ANOMALY DETECTION PERFORMANCE OF DSTCL-ATG UNDER DIFFERENT NUMBERS OF CROSSFORMER ENCODER BLOCKS

α	<i>FAR</i>	<i>FDR</i>	<i>gmean</i>
1	0.0350	0.9935	0.9791
2	0.0456	0.9821	0.9682
3	0.0575	0.9800	0.9611

DSTCL module and statistical analysis of time consumption, whose results are presented in Tables VII and VIII, respectively. It can be seen that the detection performance and time consumption of all methods are similar, and the differences are almost negligible. This indicates that ESP anomaly detection is insensitive to different prediction or reconstruction strategies. Notably, single-point prediction is inherently simpler; unlike multistep prediction, it does not require consideration of step selection, and when compared to full-window reconstruction, it demonstrates superior overall performance. Therefore, our framework adopts single-point prediction as it strikes an optimal balance between model simplicity and detection performance.

G. Hyperparameter Selection

Model performance can be influenced by adjustments to the model's hyperparameters. In this part, we examine the the following effects: 1) the number of Crossformer encoder blocks α ; 2) the dimension of embedded sequences over time Td ; 3) the number of decision trees nt ; and 4) the depth of each decision tree dt .

1) *Effects of the Number of Crossformer Encoder Blocks*: Given the significance of ESP spatiotemporal feature extraction, it is essential to investigate how the number of Crossformer encoder blocks α affects detection outcomes. The influence of different block numbers on the detection performance is investigated, and the results are shown in

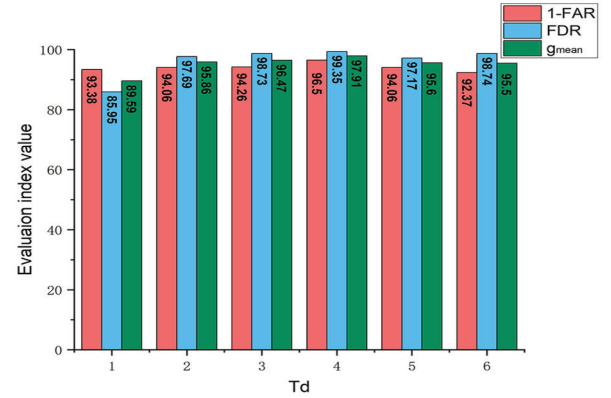


Fig. 5. Anomaly detection performance of DSTCL-ATG under different dimensions of embedded sequences over time.

Table IX. It is evident that a single Crossformer encoder block can achieve optimal performance; however, as the number of blocks increases, performance tends to decline. This indicates that the overall performance of the model can be improved by judiciously selecting the number of Crossformer encoder blocks

2) *Effects of the Dimension of Embedded Sequences Over Time*: The impact of the embedded sequence dimension over time Td is explored, with results depicted in Fig. 5. It can be observed that the model achieves optimal performance when Td is set to 64. Furthermore, variations in detection outcomes are minimal when altering Td , indicating that the proposed method exhibits a robust insensitivity to the selection of Td .

3) *Effects of the Number of Decision Trees*: The impact of the number of decision trees nt in the random forest regressor on detection performance is examined. The corresponding detection results for varying numbers of decision trees are presented in Fig. 6. Notably, the model utilizing 100 decision trees achieves the highest g_{mean} . Conversely, when nt is set either too low or excessively high, a marked

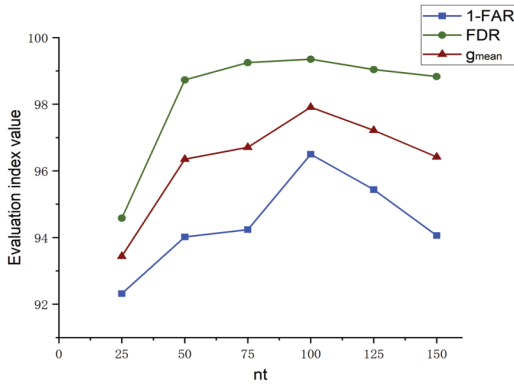


Fig. 6. Anomaly detection performance of DSTCL-ATG under different numbers of decision trees.

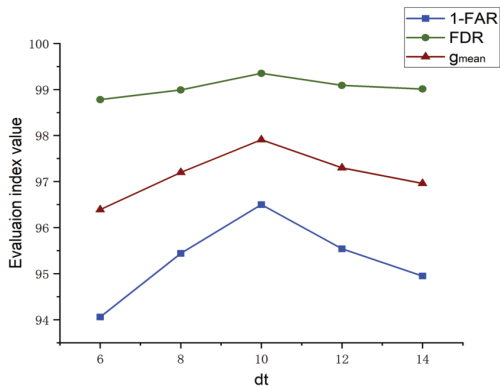


Fig. 7. Anomaly detection performance of DSTCL-ATG at different depths of decision tree.

decline in performance occurs. This observation underscores that optimizing the number of decision trees nt can enhance model performance.

4) *Effects of the Depth of Decision Tree*: The impact of the depth dt of each decision tree in the random forest regressor on the detection performance is studied. Fig. 7 displays the detection results with different depths of decision trees. It is noteworthy that the model achieved the best detection performance when the depth of each decision tree is set to 10. On the other hand, the performance shows varying degrees of degradation when the dt is set too low or too high, indicating that choosing the appropriate dt can improve the detection performance.

V. CONCLUSION

This article proposes a novel DSTCL-ATG method for ESP anomaly detection. To effectively model the complex dynamic behaviors of ESP during normal operation, the DCL architecture is employed, and the Crossformer encoder is integrated to capture the spatio-temporal coupling characteristics of ESP data, enhancing the modeling performance. Furthermore, a random-forest-regressor-based ATG strategy is developed, enabling real-time generation of adaptive thresholds to mitigate false alarms caused by fluctuations in ESP states. The proposed method is rigorously validated using

real-world data from four distinct ESP failure wells, demonstrating its effectiveness and superiority. Looking ahead, future research will focus on advancing ESP anomaly detection methods that are driven by both data and domain knowledge. For instance, domain knowledge is incorporated into the optimization function of the contrastive learning framework via a regularization mechanism. This integration facilitates the influence of mechanistic knowledge on the learning process of the data-driven model. The robustness and generalization performance of the model can be assessed through experiments conducted in electric pump wells across various oil fields.

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